

WAJOUUD H NOORANI

👤 Born on 24/04/1997 ✉ wajouudnoorani59@gmail.com 🌐 Wajoud Noorani 🌐 Wajoud Noorani
☎ +91 9590010102 📍 Belgaum, Karnataka, India, 590001

Education

Manipal School of Information Sciences, MAHE, Manipal

ME(Master of Engineering) in Big Data and Data Analytics 8.37/10.00 2019 – 2021

Manipal Institute of Technology, MAHE, Manipal

B. Tech (Hons) in Mechatronics Engineering 7.55/10.00 2016 – 2019

Profile Summary

- Innovative Data Scientist with a strong background in application development, advanced algorithm design, and neural network architecture. Proficient in a variety of programming languages, passionate about pioneering cutting-edge data science solutions to address real-world challenges..

Programming Skills

- Languages: **Python**, R Program , SQL, MATLAB
- OS Platform : **Linux**, **Windows**, **macOS**
- Software Tools : **Pytorch**, Tensorflow, Numpy, Pandas, Keras, Matplotlib, OpenCV
- Cloud Platform: **AWS**, E2E, GCP
- Platforms: **VS Code**, Jupyter Notebook, Google Colab
- Database: **NoSql** such as MongoDB, RockDb, Pickledb, PostgreSQL

Work and Research Experience

- 2023, Feb – Present | Data Scientist at **JAR**
 - **Spends Tracker** | Engineered a comprehensive real-time expense tracking system that leverages Natural Language Processing (NLP) modules and regular expressions for efficient SMS data extraction through Kafka. Implemented a Naive Bayes classifier for SMS categorization (spam/transaction) and incorporated a neural network to further enhance accuracy. Developed a bank-specific text extraction method for precise identification of expenses. The system extracts relevant information such as amounts and balances, providing users with a detailed overview of their total expenditures.
 - **Fraud Detection using graph Database** | Engineered an advanced fraud detection system by integrating Neo4j graph database technology. This module utilizes a holistic approach, considering various factors such as IP location, purchase patterns, VPA check, KYC verification, name matching, and gold buying/selling history. Each of these factors is represented as a node in the graph database, establishing connections for a comprehensive analysis. The system computes a weighted sum, effectively determining the probability of an individual's potential involvement in fraudulent activities. This innovative approach enhances accuracy and responsiveness in identifying and mitigating fraud across diverse scenarios.
 - **NLP on SMS Data** | Created an SMS classification module for real-time categorization into spam or transaction messages. Utilized a Multinomial Naive Bayes classifier with text vectorization, incorporating 61 feature extractions. Employed custom regex and Python functions for sentiment analysis, predictive analytics, and amount recommendations, addressing specific business requirements."
- 2021, Jul - 2023, Jan | Associate Data Scientist at **Happymonk AI Labs Pvt Ltd**
 - **P2P Data Flow with Pub/Sub** | Developed a comprehensive data communication system utilizing the Onvif protocol for locating surveillance systems and integrated it with a Gstream pipeline for efficient data retrieval via NATS Jetstream. Additionally, designed an asynchronous data pipeline for synchronized data from multiple cameras, with a focus on image processing using Multiprocessing. Integrated detection and activity modules into a unified batch flow for live camera feeds, ultimately generating JSON metadata for web and mobile applications.
 - **Pothole Detection for Telangana Mobility AI Grand Challenge** | Developed a robust pipeline for handling DD Pie and GoPro videos, extracting metadata including time-stamps and geo-locations. Implemented frame processing through a road classification module (AlexNet) and a pothole detection module (Detectron2), culminating in the creation of a comprehensive JSON data structure
 - **Streaming Surveillance Face Recognition**
 - * Engineered a Java-based front-end face detection module, enabling the extraction of 128 facial landmarks. This data was then communicated to the face detection module for comprehensive analysis, encompassing facial features, emotions, and age recognition.

- * Developed a robust face detection module proficient in determining a person's presence through CID correlation. Seamlessly integrated secure key-value data storage for precise person re-identification.
- * Orchestrated a comprehensive face re-identification process, encompassing CID extraction, image analysis, comparison, and result generation, ensuring a cohesive and effective face recognition system for streaming surveillance.
- **Activity Detection** | Utilized PyTorch Lightning Module in conjunction with a 3D ResNet architecture and the Kinetics human video dataset to train the activity module. Incorporated a deep Residual network for image recognition. Successfully integrated this module into a multi-camera pipeline operating asynchronously alongside other modules
- **Object Detection with re-identification** | Developed a multi-object detection model using PyTorch with a leaky ReLU, sigmoid activation, and binary cross-entropy loss. Implemented Mahalanobis distance for object re-identification in surveillance systems.
- **Saree Detection with AI** | Created a CNN for saree color and border detection, optimizing with the Adam algorithm. Enhanced functionality by integrating NATS Pub/Sub for peer-to-peer communication.
- *2020, Aug – 2021, Jul* | Research Inter at **Nokia** Networking Pvt Ltd
 - Contributed to a 5G Research and Development Team, specializing in the creation of an internal Display module. This module processed diverse network parameters from various sub-modules, analyzed the data, and presented it based on network setting protocols.
- *2019, Jan – 2019, May* | Research intern at Indian Institute of Science (**IISc**)
 - Designed and implemented a micro control unit using an Arduino chip for regulating potato plant growth in an Aeroponics chamber. Integrated various sensors, including DTH (Temperature and Humidity), Pressure, and Analog Actuator sensors, and conducted data analysis to optimize water consumption and system efficiency.
- *2015 -2021* | Other Projects
 - *ME(Master of Engineering)* | Academy Project
 - * **Travel-Optimization** | Designed optimized route planning solution using genetic algorithm, forecasting travel times and selecting the best routes through iterative specimen breeding for enhanced efficiency.
 - * **Car-Lane-Change-Detection** | Develop a multi-module system for lane detection, vehicle/obstacle recognition, and collision risk assessment
 - **Stock-Market-Analysis** | Contributed to a project analyzing live stock market data for companies like HDFC, TATA Motors, ICICI Bank, and Infosys. Leveraged Keras and TensorFlow for data analysis and successfully predicted the closing stock value of TATA Motors.
 - **Recommendation-System** | Led a diverse recommendation system project with three distinct methods. The first involved weighted averages derived from overall movie popularity. The second utilized content-based recommendations using metadata and TF-IDF vectors with a linear kernel. The third provided personalized recommendations based on user preferences.

Other Work Experience

- Knowledge about **Web and Mobile Development** environment such as React Native, GraphQL queries and schema
- Knowledge and work experience on **Nvidia Development Board** such as **Nvidia Jetson Series**
- Knowledge and work Experience on **Microcontroller** | **Microprocessor kits** such as **Arduino, Raspberry Pi**
- Knowledge about basics **Networking** concepts such as Network Topology, Networking devices, Open Systems Interconnection
- Knowledge about basics **Docker** concepts like docker engine, docker image, docker container.

Achievements, Awards and Scientific Recognition

- *2023* | Achieved second runner-up position at the Unified Logistics Interface Platform (Logixtics) Hackathon 2022, organized by the Government of India, for activity detection and human identification.
- *2019, 2021* | Elected as a Student Council member during my postgraduate studies at the departmental level.
- *2016* | Participated in the National Level Technical Fest, Project Titled - Water Robot.
- *2012* | Was a cadet of the National Cadet Corps (NCC).
- *2010* | Gold Topper for 21th regional Abacus Competition.
- *2010* | Silver Topper for 13th National Abacus Competition.
- *2009* | Won the Third place in district level swimming competition (4*50 freestyle).